

Investigating the Potential of User Interface Shortcuts and Control Panes to Support Mobile Phone Use by People with Mild Dementia – a Diary Study

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ABSTRACT

Given the necessity of mobile phones to participate in everyday life and past research indicating the inaccessibility of mobile phones for people with dementia, there is a need to design more accessible mobile interactions to ensure their full inclusion in society. We conducted a two-week diary study with three people with mild dementia to investigate their experiences with and perceptions of user interface (UI) shortcuts and control panes. Findings from this study demonstrate the potential of UI shortcuts and control panes to address specific mobile phone accessibility challenges people with mild dementia experience. This work provides a technical contribution in the form of control panes for mobile phones, as well as design considerations to improve the viability of mobile phone shortcuts and controls panes as future assistive technology for individuals with diverse cognitive abilities.

CCS CONCEPTS

• **Human-centered computing** → Accessibility; Empirical studies in accessibility; Accessibility; Accessibility systems and tools.

KEYWORDS

Dementia, Cognitive Accessibility, Mobile Phones, User Interface Design

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1 INTRODUCTION

Researchers have shown an uptake in mobile phone use [15, 23] and preference for mobile phone use over other devices [12, 38] by

individuals living with mild dementia – a term used to describe a set of neurodegenerative disorders which involve changes in cognition, affecting one’s ability to engage in daily tasks and activities [31]. Despite this increase, there remain open accessibility challenges for this population, such as: typing on mobile phones due to small key size [10]; filtering which apps to download and use [19, 20]; navigating to apps and features due to the vast amount of apps and information available [11]; and executing all steps of multi-step tasks on their mobile phones in moments of time-pressure, high stress, or fatigue [11].

Technological innovations have been developed to simplify mobile phone user interfaces (UI)s, such as Elder Launcher [17] and Simple Mode [33], which launch apps and call favorite contacts through a simplified home screen launcher. Also, senior-friendly phones, such as the Jitterbug smartphone [40], which take simplification one step further than Android Launcher apps by providing simplified UIs for built-in apps (e.g., phone and camera). But not all people with dementia are also older adults [35], which may affect their adoption of these systems. An alternative for UI simplification is to use built-in “virtual assistants” to create “routines,” making it possible to perform a sequence of actions at one time. With shortcut apps (e.g., Siri Shortcuts on iOS [41], routines in Google Assistant for Android [42]) users can place icon-style shortcuts on their home screen to start a routine quickly with one-tap on the shortcut. Yet, these icon-style shortcuts may not be accessible to people with dementia due to lack of familiarity in the design of the non-customizable icons [25, 32, 38].

In this study, we sought to investigate the potential of a prototype UI personalization application in addressing some of these mobile phone accessibility challenges. The prototype included two distinct interfaces: 1) Action Blocks is an application that provides a personalized home screen with shortcuts that enable one-tap access to digital services on Android devices; and 2) Action Shortcuts provides an intermediate interaction layer designed to be a control pane that automatically changes with different types of apps to assist users in navigating follow-on actions within an app. Through a two-week diary study with three people living with mild dementia, this study demonstrates the potential of UI shortcuts and control panes to address specific mobile phone accessibility challenges people with mild dementia experience. Additionally, this study provides a technical contribution in the form of control panes for mobile phones and design considerations to make mobile phone

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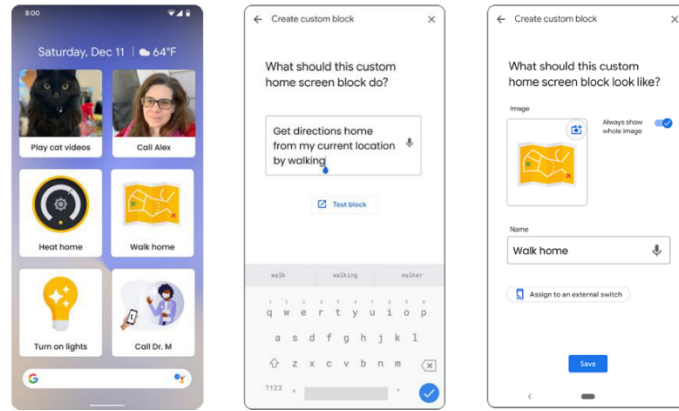


Figure 1: (left): An example of customized, one-tap blocks appearing on an Android home screen. (middle): Entering a string command to Assistant as part of the block set up process. (right): Customization options – selecting an image and creating a custom name.

shortcuts and control panes a viability future assistive technology for people living with mild dementia.

2 SYSTEM DESIGN

2.1 Action Blocks

Action Blocks is an Android app intended to make mobile phone or tablet use easier by adding customizable home screen buttons (as Android widgets) that can initiate favorite activities, such as calling a friend, watching a predefined movie or playlist, checking in with support staff, and navigating home by walking, with a single press (Figure 1, left). Each of these one-tap buttons is assigned to a customizable Google Assistant command (e.g., “Play cat videos on YouTube”, “Call Mom”, etc.), which is set up by users when they create the button. Tapping a button creates an Android Intent with the associated command and sends the Intent to Google Assistant through an Assistant API [43]. Once Assistant receives the Intent, it opens the Assistant app and executes the input command as if it were spoken out by the user. For example, a “Call Mom” command triggers Assistant to open the Dialer app, passes Mom’s contact information to it, and dials Mom’s number. When configuring a home screen button, users can set it up to perform any Assistant command (Figure 1, middle). They can assign a personal photo, illustration, or familiar symbol to the button, and resize it (Figure 1, right) to facilitate easy identification of these one-tap buttons. Please refer to our past work [7] for further details on the rationale for the Action Blocks design decisions.

While Google Assistant is a powerful tool and can help people perform a variety of actions on their phone, its functionality is mostly limited to simple, one- or few-step actions, and it cannot complete complex, multi-step tasks that require user input (e.g., open the Gallery, select a particular picture, and share it with a family member). These limitations of Google Assistant also limit what Action Blocks can do. While Action Blocks makes it easy to initiate a simple task, in the case of a complex task, the user must navigate a sequence of additional UI elements. For example, state changes, such as a user being logged out of their account due to

inactivity or forced authentication by an app service provider. In these scenarios a user is presented with an authentication screen that could cause confusion, pose challenges for entering authentication details, or even provoke alarm around perceived account security (as described in past work [26, 27]). Therefore, while Action Blocks can greatly reduce the cognitive burden of initiating user journeys and specifying initial parameters, it does not provide continued support for users to interact with the app and complete their task, once Assistant has launched them into the app. This design challenge was uncovered through user feedback in [7], which we respond to with the design of Action Shortcuts, described below.

2.2 Action Shortcuts

To address the known limitations of Action Blocks, we propose Action Shortcuts to provide users with continued support to complete tasks that require user input. It aims to provide a consistent, simplified intermediate interaction layer for users to navigate follow-on actions in an app, after Action Blocks has launched them into it. The need for Action Shortcuts comes from two sources. First, all apps are different, and therefore, even common functionalities (e.g., Confirm and Cancel) can look very different from app to app, making them hard to learn and discover for people with diverse cognitive abilities. Consider YouTube (Figure 2 left image) and YouTube Music (Figure 2 middle image), both apps have a ‘Pause’ function as media players, but they are visually distinct and located in different areas of the UI. Second, even within a single app, the location, size, and appearance of a control may be different in different instances of launching the app (e.g., changes to app UI occurring after an update).

Action Shortcuts includes a floating window (an accessibility overlay) on the right edge of the screen, which we refer to as the control pane (a synonym for control panel), that contains uniform app controls for users to interact with the underlying app (see Figure 2). At the time of this study, the prototype provided in-app controls for three selected apps: YouTube, YouTube Music, and Duo. These apps were selected because they are popular, free applications supporting common mobile phone tasks: media consumption and

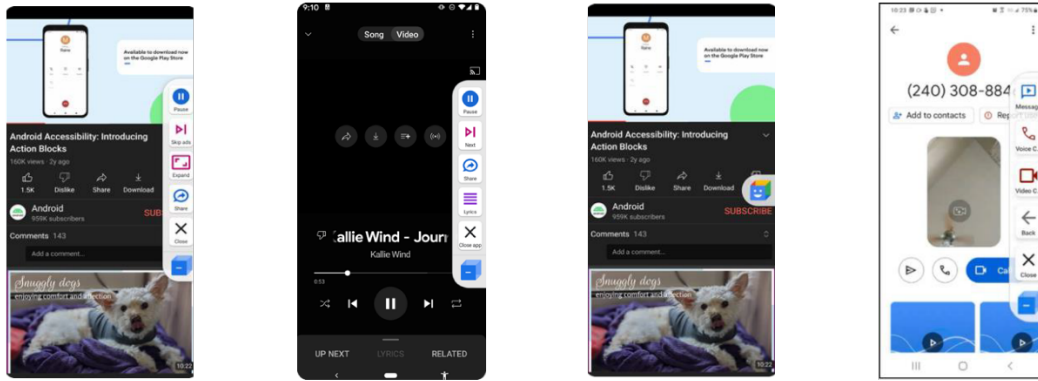


Figure 2: Action Shortcuts control pane for YouTube (left), YouTube Music (middle-left), in a minimized state (middle-right), Duo (right).

communication. App controls are grouped into multiple control sets that correspond to different app functionalities (e.g., media controls, calling controls, etc.). Controls of the same functionality set are always presented together on the control pane so that users have a more consistent and predictable interface, reducing cognitive load by creating familiarity of control placement and appearance. Apps are assigned to a control set based on what shortcuts make sense for and are likely to work within the app. Each control in the control sets (except for the “Close” control) is a “shortcut” mapped to a UI element on the screen. When a user taps a control, Action Shortcuts performs a click on the underlying matched UI element through a dispatched gesture on the element. The ‘Close’ control takes users to the Home screen by performing the global accessibility action to go Home, thus does not reference any UI element. It is added to all control sets to provide an efficient way for users to exit the app. Users can minimize the control pane by tapping the blue collapse icon on the bottom of the pane. When the pane is minimized, only the Action Shortcuts icon remains on the screen (Figure 2, middle-right image). Tapping this icon expands the control pane. The intermediate interaction layer was made always visible but minimizable to take up less screen space so that the user could still access the full functionality of an app. This is in line with past work, which describes the importance of pinning UI elements to specific locations on the screen to provide quick and easy access for people with dementia [38].

Because all controls (except for the ‘Close’ control) need to reference a UI element on a specific screen, we developed an algorithm to identify what controls are available in an application interface and find the underlying UI element associated with each of these controls. Action Shortcuts uses Android Accessibility APIs [2] to obtain the properties of each UI element on the screen. It then applies a set of heuristics to determine if a UI element matches a control by determining whether the resource id, text or content description of a UI element contains keywords associated with the control. In case of an app update where the predetermined properties no longer work for the updated app interface, we also used an icon type detector that is based on the CenterNet architecture [39], to detect the icon type of the UI element based on the UI screenshot,

and used this information, along with other heuristics, to determine if a UI element matches a control. If no matched UI element is found for a control in the control set for the app, the control is disabled and grayed out on the control pane. When the reference element on screen is hidden initially (e.g., the pause button on the YouTube video player in Figure 2, left image), a two-step Macro is executed after the ‘Pause’ control is clicked: (1) click the video player to show all video control buttons, and (2) click the ‘Pause’ button to pause the video. We created a data structure to serialize information about the clicks to be performed by the Macro and wrote scripts to detect conditions when a macro should be executed and identify UI elements and perform clicks on them step-by-step.

2.3 How The Two Interfaces Work Together

Action Shortcuts is an accessibility service built on top of Action Blocks. Figure 3 demonstrates how the two interfaces interact and work together. First, Action Blocks is used to create a home screen button to perform a task (e.g., play cat videos on YouTube). After the user clicks the button, Action Blocks starts the task by opening the desired app. Action Shortcuts monitors screen updates and notices that a supported app is opened, it presents that app’s control set on the pane. It was important that the two interfaces be tested at the same time as each interface addresses a different accessibility barrier to mobile phone use for people with dementia identified in previous literature, 1) difficulty navigating to apps and features due to the vast amount of apps available [11]; and 2) difficulty executing all steps of multi-step tasks on their mobile phone in moments of time-pressure, high stress, or fatigue [11]. Therefore, we intended for these two technologies together to make Android more accessible by: (1) making interacting with the UI easier through large, customizable shortcut buttons that are easy to recognize through Action Blocks; (2) helping users understand the UI better, by presenting controls linked to actions that are likely to be performed on each screen through Action Shortcuts.

3 METHODS

We conducted a two-week diary study with three participants with mild dementia. Below we describe our recruitment procedures,

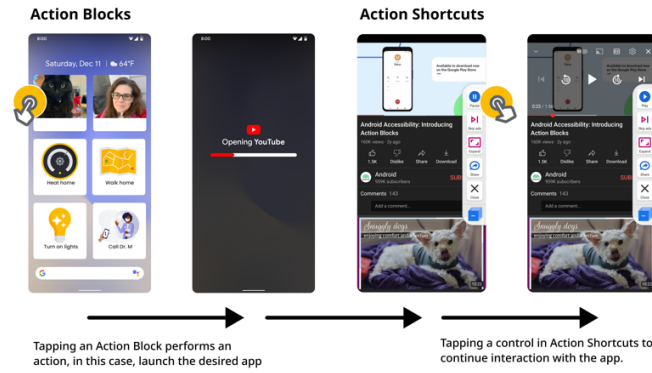


Figure 3: Illustration of how Action Blocks and Action Shortcuts work together.

participant demographics, as well as data collection and analysis methods.

3.1 Recruitment and Participant Demographics

Recruitment was completed through convenience sampling. To participate in this study, individuals had to self-report a clinical diagnosis of mild dementia and use an Android mobile phone – which was necessary to download the prototypes. We did not collect information on how long participants had used an Android phone. All three participants (2 female, 1 male) resided in the United States, identified as white, had some college education but no degree, and were within the age range of 55-74. None of the participants in this study reported receiving assistance from others with their mobile phone use, indicating at least passing knowledge and experience with mobile phone use. No participants reported prior experience with accessibility launchers.

3.2 Data Collection and Analysis

Data collection included 3 one-hour semi-structured interviews with each participant as well as two weeks of diary entries collected via a Google Form and voice memos. The study was completed remotely and all synchronous interactions with participants were via Google Meet and were audio/video recorded, where participants joined the call both from their laptops and their mobile phones, sharing their screen on their mobile phones. Before the study began, participants completed electronic informed consent forms and provided demographic information. For the diary entries, participants were asked to reflect on what they did or did not enjoy about using the prototype that day, challenges they experienced, and miscellaneous feedback. We had access to participants' diary entries as they were submitted, which informed the interviews. We did not ask participants to complete specific tasks with the prototype every day. In the initial interview we walked participants through downloading the prototype, described how both Action Blocks and Action Shortcuts worked, assisted each participant in personalizing their Action Blocks, and gathered participants initial feelings and perceptions of the prototype. Upon initial download, three pre-made Action Blocks were automatically placed on participants' devices: (1) a YouTube video describing how to use Action Shortcuts, (2) a

Google Form for daily diary entries; and (3) a Duo video message to the researchers to record daily feedback. One week after the initial interview date we conducted the mid-way interview where we addressed any technical difficulties, asked clarifying questions concerning diary entries, gleaned their overall feelings and perceptions towards the shortcuts and control pane, and made any requested changes to their personalized home-screen, Action Blocks, if they were not able to make those changes themselves. One week after the mid-way interview we conducted final interviews, clarifying questions about participants' overall feelings and perceptions towards UI shortcuts and control panes and their perceptions of its usefulness. Following the interview, participants uninstalled Action Shortcuts but kept Action Blocks as it was publicly available on the Google Play Store prior to the study. Participants were given a \$275 gift card as a thank you for their time and participation.

In total we collected 9 hours of interview data and 33 diary entries, including: 28 Google Form entries and 5 Duo voice messages, which were approximately 1-2 minutes each. Each voice message was recorded and transcribed by the first author. We used reflexive thematic analysis inductively [6], where the codes and themes developed in this analysis were grounded in our data. Separate analyses were conducted for each interface. The themes developed are italicized in the finding section. Influenced by critical disability theory [13, 28], the aim of this analysis was to uncover 1) the voices of people with mild dementia towards UI shortcuts and control panes and 2) practical challenges they may encounter. None of the research team identified as a person living with dementia. Further, the research team is training in ability-based [37] and inclusive technology design [16]. Our theoretical alignment as well as our positions and training informed our analysis.

4 FINDINGS AND IMPLICATIONS

Below we outline findings relating to the UI shortcuts and control panes, the implications of these findings for the design of future mobile phone assistive technologies for people with dementia, and limitations/future work.

4.1 User Interface Shortcuts - Action Blocks

Action Blocks *helped participants to multi-task*, using their mobile phone while simultaneously engaging in everyday activities, which clinical research has shown can be especially difficult for people with dementia [18]. For example, Sean described: “I really love the app. . . During the day, if I’m doing housework, I’m doing the dishes, I can just hit that one and it plays 80’s pop on YouTube Music.” The single level menu structure of Action Blocks was helpful in *decreasing the perceived amount of time participants spent searching for content*, similar to previous work with people with traumatic brain injury [8]. As Cait described: “I do a lot of searching for YouTube videos... There’s so many things, which creates a whole bunch of searching and time.” But with Action Blocks “it shrinks YouTube down to your interests instead of the big Wide Web” [Cait]. Further, the single level menu structure *assisted participants in moments of fatigue and stress*, which previous work has found to be an inhibitor of mobile phone use for people with dementia [11]. Sean describes how without Action Blocks “If I needed to get ahold of [my wife] urgently because I was having a stressful situation, it’s like four or five steps and sometimes it’s overwhelming.” With the shortcuts Sean can “use the 1 touch to call my wife. . . I mean that was just so simple and I love the fact that it’s got that one punch button for [calling her].” Given these benefits, participants described how they used the shortcuts more over the two weeks. And, as Gwen described “the more I use them, the more I find out about how to use them,” referring to how she continued to add more shortcuts throughout the two-week study.

All three participants described the *need to simplify the initial setup process of the UI shortcuts so that they could set it up without assistance from others*: “Somebody like myself could not set that up... You had to walk me through it. So, I think that’s where the challenge is going to be, where they’re going to have to have a caregiver, or somebody that’s kind of a little bit phone or computer savvy to help set them up and customize things” [Gwen]. One specific challenge participants encountered with setup was *difficulty resizing and moving the custom one-tap blocks*, which aligns with past work that found precise touch screen movements can be difficult for people with dementia due to changes in dexterity [9]. To address this challenge, participants wanted automated sizing of the one-tap block style by letting users choose from a list of potential styles. A second challenge participants described with initial set-up was *managing multiple pages of apps on their phone*, this aligns with past work which found that people with dementia have difficulty managing multiple windows on desktops [38] and filtering through content on their phones [11]. To address this, participants suggested all personalized one-tap shortcut blocks be automatically placed on a single page, specifically the second page next to the home screen, rather than adding new pages to place new blocks. Finally, when creating the customized shortcuts, participants had *difficulty wording the commands for Google Assistant*. As Sean reflected on his experience, he described how “when we were setting it up one time, it took us three or four times to get the right verbiage. . . If you’re not specific enough, you’re not going to get the detail that you’re looking for. A lot of it is a verbal command that you have [to use] the exact precise words in there.” This difficulty may be in-part due to changes in pragmatic language abilities – social rules

of language for the purpose of communication – that people with dementia often experience [29], demonstrating the need for future systems to follow guidelines developed for voice-user-interfaces for people with dementia [34].

4.2 User Interface Control Panes – Action Shortcuts

Providing a control pane that could be accessed in the same location regardless of the application being used helped to *address challenges participants experienced with executing all steps in multi-step tasks*, as described in past work with people with dementia [11]. Gwen describes her frustration with using apps without the control pane: “I swear on every app you’d have to find all those things and then try to remember all those things”, referring to executing tasks within various apps. Instead “having the shortcuts, you just have to remember Action Shortcuts once you bring up the app. So that’s really nice” [Gwen]. This ease in executing multi-step activities may be at least in part due to *the design of the control pane being perceived as comforting*, where Sean comparing it to the design of “an old tape recorder, which gave me comfort because it wasn’t something completely new.” This finding, similar to past research on retro designs in technologies for people with dementia [14, 36], may be indicative of legacy bias, where users tend to prefer interactions from existing technologies as this may lessen the load on the mind and make new systems easier to adapt to [4, 21].

Further, participants reflected on how the control panes expanded their technical repertoire, addressing a frequently cited barrier to technology use for people with dementia [22]. For example, when using the control pane on YouTube Cait was “able to share a workout video with gal pals,” which she previously did not know how to do. Additionally, participants found the control pane to be “very convenient,” particularly enjoying the “ability to skip music and close out quickly” [Cait] and return to the home screen – tasks that were increasingly difficult. With the overall helpfulness of the control panes, participants described several other applications on their phone they would benefit from control panes, such as audiobooks [Sean]; video-conferencing apps [Cait]; email [Sean, Gwen]; phone calls [Sean, Cait]; calendar apps [Gwen]; and the camera app [Cait].

Yet participants expressed a clear *need for the customization of how and when the control pane is displayed*. Sean felt the persistent, immovable icon on the right side of the screen that expanded into the control pane was “in the wrong placement. It’s in the way when I text or reply to someone on Facebook.” It also interfered with Cait’s ability to “scan my concert ticket. We tried sliding the ticket around, but nope.” When asked if they would prefer to not always have the icon visible, Cait responded: “I don’t think it should ever go away because out of sight out of mind.” Similarly, Sean felt the icon should be always visible but to “make it so I can move it [the Action Shortcuts icon] or drag it around for better placement.” In contrast, Gwen felt that the control pane icon should “come up automatically with the apps that use it rather than be there all the time.”

Additionally, we discovered there was a *limit to the acceptable amount of adaptability of control panes* for them to remain accessible to participants, aligning with past work that found adaptive

UIs are not always desirable for all user groups [24]. Cait described how using the control panes with Duo was difficult because “these buttons are changing a lot. . . with every click,” referring to the control set changing from the calling control set to the recording control set. This change in control created uncertainty for participants who weren’t sure if they were recording a voice message or in the middle of a call. To address this challenge, participants wanted a deep structure when navigating within apps with multiple functions. For instance, the menu after selecting the block to call someone “should be compressed into one page, two choices” either “leave a message” or “video call” [Cait]. This would essentially create a sandbox UI with an initial menu where users could select their intended function within an app before entering the app, where only the intermediate interaction layer that fit their selected activity would be displayed.

5 LIMITATIONS AND FUTURE WORK

Although the number of participants in this study aligns with the sample sizes of previous accessibility focused diary studies [1, 5, 30], the small sample size and the lack of diversity in participants attributes (e.g., one group of users with diverse cognitive abilities—people with mild dementia) limit the generalizability of the findings. In future work we aim to test Action Blocks and Action Shortcuts with a more diverse set of participants, including those with different cognitive abilities. Another limitation of this study is that we did not collect data on participants’ mobile phone use prior to the study to conduct a comparative analysis of how mobile phone use may change with the uptake of Action Blocks and Action Shortcuts – a direction we plan to take in future work. Finally, findings uncovered in this study demonstrate the need for future iterations of the prototypes to include more automated personalization and expand the control pane to additional applications. With the use of well-trained deep learning models for screen understanding (e.g., UIbert [3]), in future work we aim to expand the control sets Action Shortcuts can offer to support a wide variety of apps through a more intelligent adaptive UI.

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